# ICNNA Architectural Refactoring
# Phase 3 Minute — Roadmap Definition

From v1.4.1-beta to v2.0.0: twenty-eight sub-releases across four arcs

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**Document Log:**

- 14-Apr-2026: FOE/Claude — Document created. Records Phase 3 (roadmap definition) in full. Twenty-eight sub-releases organised in four arcs plus eight cross-cutting threads. Phase 3 formally closed by this minute.

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## 1 Session Overview

**Date:**
14 April 2026

**Participants:**
Felipe Orihuela-Espina (FOE), Claude Opus 4.6 (Computational Research Architect role).

**Predecessors:**
Phase 1 minute; Phase 2a minute (12-Apr-2026); Phase 2b minute (13-Apr-2026); Phase 2c minute (13-Apr-2026); Phase 2d closing minute (14-Apr-2026).

**Objective:**
Produce the concrete versioned roadmap from ICNNA v1.4.1-beta (current) through v1.4.2 (the "modernisation mode begins" milestone) through subsequent sub-releases to v2.0.0; slot the sixteen deferred decisions (D.1–D.16) from Phase 2d §3.C; identify and fold seven additional items surfaced during Phase 3 review.

**Outcome:**
Roadmap defined. Twenty-eight sub-releases identified across four arcs (Readiness, Spine+Experimental-design, Peripheral+Behaviour, Endpoint). Eight cross-cutting threads embedded. Session budget estimate: **∼65–95 sessions** across ∼2–3 years at research pace. All sixteen D.∗ items slotted. Seven new items incorporated (Phase 3 additions, §3).

**Model used:**
Claude Opus 4.6.

**Status:**
**Phase 3 is formally closed with this minute.** Phase 4 (execution planning: v1.4.2 session-level breakdown) to follow in a separate session.

## 2 Phase 3 Inputs and Starting Posture

Phase 3 opened with the canonical Phase 2 closing artefacts:

- The nine design principles P1–P9 (Phase 2c minute §4.2).
- The seven operating rules for internal transition (Phase 2d minute §E.1, Rules 1–7).
- The six coexistence policies with Cantor (Phase 2d minute §E.2).
- The six handover policies (Phase 2d minute §E.3).
- The five communication policies (Phase 2d minute §E.4).
- The open-items registry (Phase 2d minute §F), particularly the sixteen deferred decisions D.1–D.16.
- The v1.4.2 pre-packed milestone content (Phase 2d minute §E.1 Q4): test infrastructure, GUI deprecation, `+icnna.compat` creation, $\gamma \to \beta$ migration of 5–6 sites, R2021a declaration.

FOE supplied the Phase 3 prompt at session start, taking precedence over the draft at `claude/prompts/Phase3_context_prompt.txt` per the Phase 2d §G.4 instruction.

Phase 3 proceeded iteratively: (i) a first-draft roadmap presented for review; (ii) five substantive gaps flagged by FOE (experimental-design flexibility, registration-tool behaviour, QC behaviour, signal processing, UML/docs); (iii) revised roadmap; (iv) seven further items surfaced in a "what am I missing" sweep; (v) confirmations; (vi) this minute.

## 3 Phase 3 Additions Beyond Phase 2d Deferred Decisions

Seven items were surfaced during Phase 3 review that were not explicit Phase 2d D.* items but have non-trivial implications for the roadmap:

| ID | Item | Home |
|---|---|---|
| N.1 | **Experimental-design generalisation**: ICNNA's `experiment/session/condition` model currently struggles with event-related designs, within-group allocation, and multifactorial designs (users "abuse" the session concept). Generalisation is a survival-tier requirement of the `experiment` redesign. | v1.4.7a (memo), v1.4.7b (impl) |
| N.2 | **Registration behaviour**: The registration tool is not a channel-LocationMap wrapper but carries genuine computational behaviour (standards alignment 10/20/10-10/10-5, atlas alignment, mesh, distances). Research drift expected; FOE to supply material. | v1.5.2 (standards), v1.5.3 (atlas+research) |
| N.3 | **QC concrete behaviour**: `+icnna.data.core.qc` and `qcCriterion` exist as classes but lack concrete criterion implementations. Split tiered: survival tier = current ICNNA criteria (saturation, mirroring, integrity); enhancement tier = SCI, CV, and others from overprocessing/pipeline-optimisation literature. | v1.5.4 (survival), v1.6.4 (enhancement) |
| N.4 | **Signal processing expansion**: Current `+icnna.op` is thin. Serious enhancement needed in motion correction (beyond TDDR), physiological regression (SPA-like, beyond short-channel), and synthetic data generation (absent). Three dedicated sub-releases. | v1.6.0, v1.6.1, v1.6.2, v1.6.3 |

| ID | Item | Home |
|---|---|---|
| N.5 | **Documentation modernisation**: UML framework decision (keep `m2uml`, retire ArgoUML, adopt PlantUML); User Guide LaTeX migration (currently `.docx` lineage); Limitations document folded into User Guide appendix. | v1.4.3.3 (scaffolding), cross-cutting thread |
| N.6 | **User `.mat` backward-compatibility**: When `oo/@X` classes are deleted at v1.8.0, users' saved `.mat` archives containing legacy class instances will fail `load()`. Mitigation via `+icnna.compat.load<ClassName>.m` adapters shipped with every modernised class in Arc B. | Cross-cutting thread (Arc B) |
| N.7 | **Operational infrastructure**: Error-ID convention codification; MATLAB toolbox dependency declaration (Statistics accepted; Signal Processing to be avoided where feasible); CITATION.cff using DOI 10.1117/1.NPh.5.1.011011; hybrid CI (GitHub Actions on PR/-push + local coverage); bug-tracking discipline (GitHub Issues with labels + CONTRIBUTING.md + User Guide "Known limitations" appendix); performance benchmark harness (acknowledging the pass-by-value ceiling as intended, detecting only redesign-introduced regressions within that ceiling). | v1.4.3.2, v1.4.7b/v1.4.8 (benchmarks) |

These seven additions are first-class roadmap items from this point forward.

# 4 Roadmap Overview

## 4.1 Arcs

| Arc | Versions | Sessions | Theme |
|---|---|---|---|
| A. Readiness | v1.4.2, v1.4.3.1–.3 | 6–8 | Code infra + disclosure + conventions + d |
| B. Spine + experimental design | v1.4.4 – v1.4.8 | 12–17 | P1 core-class modernisation + `experiment` |
| C. Peripheral + behaviour | v1.4.9 – v1.7.0 | 27–39 | rawData, spatial, registration, QC, analysis |
| D. Endpoint | v1.8.0 – v2.0.0 | 15–25 | `oo/` removal, handover, unified GUI, User |
| **Total** | | **60–89** | |

## 4.2 Cross-cutting threads

Eight threads span versions rather than living in a single sub-release:

**T1. User Guide (LaTeX migration)**: skeleton at v1.4.3.3; chapter landings at v1.4.8 (data model), v1.5.5 (analysis), v1.6.4 (quality control); full polish at v2.0.0.

**T2. UML refresh**: initial snapshot at v1.4.3.3 (`m2uml` + PlantUML seed); spine refresh at v1.4.8; peripheral refresh at v1.5.5; final at v2.0.0.

**T3. Compat ledger drain**: opportunistic every sub-release; explicit catch-up sweeps at v1.4.8 and v1.8.0; final deletion at v2.0.0.

**T4. `.mat` load-compat adapters** (N.6): every legacy class modernised in Arc B ships `+icnna.compat.load<Cl` adapters retained past v1.8.0 `oo/` deletion; removed at v2.0.0.

**T5. Performance benchmark harness** (N.7, §6): baseline at v1.4.7b; re-run at each Arc-B release; human-read comparison; not CI-gated.

**T6. Examples gallery (cookbook)** (D.12 extension): grows organically in `doc/examples/`; FOE to contribute "series"-style worked examples as seeds when the time comes.

**T7. Bug-tracking discipline** (N.7): GitHub Issues adopted as tracker from v1.4.3.2 onward; durable known limitations mirror into User Guide appendix.

**T8. Fixture provenance** (ties to D.7 IP ledger): Claude flags source, license, and IP-owner whenever a new fixture is introduced; FOE provides ownership details; ledger updated.

### 4.3 Principle and rule traceability

The roadmap honours:

- **P1** (data model first): Arc B precedes Arc C entirely.
- **P2** (test-backed from day one): v1.4.2 lands test infrastructure; every Arc-B/C class modernised ships with tests.
- **P3** (sub-releases 1–3 sessions, each usable): honoured across all sub-releases; heavy releases (v1.4.3, v1.4.7) split explicitly.
- **P4** (GUI deprecation clean cut): v1.4.2.
- **P5** (flat architecture = design task): enforced via design-memo releases (v1.4.7a, v1.5.3, v1.9.0).
- **P6** (bottom-up, relaxed): respected in Arc B ordering.
- **P7** (survival vs enhancement): explicitly tiered in QC (v1.5.4 vs v1.6.4), signal processing (v1.6.0 survival vs v1.6.1/.2 enhancement), BIDS (export survival, round-trip enhancement).
- **P8** (research-interrupt tolerance): 1–3 session budget per release; each release leaves codebase usable; Rule 2 mixed granularity.
- **P9** (scope containment): design-memo splits prevent scope creep in implementation releases.
- **Rule 1** (soft cut): every modernised class deprecates legacy rather than deleting.
- **Rule 4** (opportunistic caller migration, cleanup at $\leq 3$ sites, D.1): absorbed into v1.4.8 and v1.7.0 cleanup sweeps.
- **Rule 5** ($\beta$ bridging via `+icnna.compat`): all bridging in `+icnna.compat/`; final deletion at v2.0.0.
- **Rule 6** (v1.4.2 is the "modernisation mode begins" milestone): locked.
- **Rule 7** (`*/*Definition` decoupled): honoured by treating paired redesigns case-by-case.

## 5 Version-by-Version Breakdown

Per-version convention: **G** goals, **M** modules, **T** tasks enumerated, **E** estimate, **R** primary risk. Cross-references to deferred decisions D.* and Phase 3 additions N.*.

### 5.1 Arc A — Readiness

#### v1.4.2 — "Modernisation Mode Begins"

Pre-packed per Phase 2d §E.1 Q4.

**G.** Land the one-time modernisation switch in a single release.

**M.** `tests/`, `+icnna.compat/`, `GUI/` (deprecation), `icnna_startup.m`, `doc/examples/`.

**T.**
1. Create `tests/` root with `matlab.unittest` skeleton and runner harness (D.15).
2. Implement `+icnna.compat.toModernTimeline`.
3. Migrate the 5–6 $\gamma$-sites in modern code to the adapter (Rule 5).
4. Deprecate `GUI/` folder: deprecation banner + release-note posture (E.4.1).
5. Enforce R2021a minimum in `icnna_startup.m` with version guard.
6. One CLI worked example in `doc/examples/` (D.12 minimum).

**E.** 1–2 sessions.

**R.** Concentrated breakage; tests-first mitigates.

#### v1.4.3.1 — "Disclosure artefacts"

**G.** Lock durable disclosure/academic artefacts.

**M.** `CHANGELOG.md`, `doc/ip_ledger.tex`, `CITATION.cff`, `doc/UserGuide/` (begun in .3.3).

**T.**
1. Seed `CHANGELOG.md` at repo root (Keep-a-Changelog), full migration from `doc/ICNNA-Version.log` (D.11, per §9 full-disclosure).
2. Collaborative `doc/ip_ledger.tex` pass with FOE's institutional input (D.7).
3. `CITATION.cff` at repo root using DOI `10.1117/1.NPh.5.1.011011`; current citation extracted from `README.md`.
4. Retire `doc/ICNNA_Limitations.doc` as standalone; content to be folded into User Guide at v1.4.3.3.

**E.** 2 sessions.

**R.** D.7 blocked on institutional consultation; advance with placeholder and revisit.

#### v1.4.3.2 — "Conventions & tooling"

**G.** Lock operational conventions and CI so later releases inherit them.

**M.** `doc/conventions/`, `.github/workflows/`, `CONTRIBUTING.md`, `ROADMAP.md`.

**T.**
1. Deprecation-warning mechanism convention (D.2).
2. Doxygen-style header template for `+icnna.compat` members (D.3).
3. MATLAB-version support statement wording (D.8) — appears in `icnna_startup.m` and User Guide.
4. Error-throwing convention document `doc/conventions/errorIds.tex` codifying current `icnna:<namespace>:<reason>` style. No behavioural change.
5. MATLAB toolbox dependency declaration `doc/conventions/toolboxDeps.tex`. Statistics Toolbox: accepted dependency. Signal Processing Toolbox: avoided where feasible; re-evaluate per-op in Arc C. Startup warning if declared dependency missing.
6. Hybrid CI setup (N.7 option c): `.github/workflows/ci.yml` running `matlab.unittest` + lint on PR/push using MathWorks' published GitHub Action; coverage computed locally, not CI-gated.
7. Bug-tracking discipline: adopt GitHub Issues with labels (`bug`, `enhancement`, `question`, `blocked-on-cantor`, `survival-tier`, `enhancement-tier`); add `CONTRIBUTING.md` describing issue-filing.
8. Commit `ROADMAP.md` at repo root (this roadmap, textual form).

**E.** 2 sessions.

**R.** CI friction on first green run; isolate by landing workflow last.

#### v1.4.3.3 — "Documentation scaffolding"

**G.** Documentation toolchain locked; User Guide LaTeX skeleton ready to grow.

**M.** `doc/UML/`, `doc/UserGuide/`.

**T.**
1. Retire ArgoUML: move `.zargo` + historical PDFs (`ICAF*.pdf`, old UML assets) to `doc/UML/historical/`; README note marking ArgoUML discarded; archive-only status.
2. Adopt PlantUML: create `doc/UML/plantuml/` with a seed flat-architecture diagram.
3. First `m2uml` run capturing current dual state (`oo/` + `+icnna/`) in `doc/UML/ICNNA/v1.4.3.3/`.
4. User Guide LaTeX skeleton: `doc/UserGuide/main.tex` with chapter stubs covering: Introduction; Installation; Data model; Sessions & experiments; Hardware support (rawData); Registration; Quality control; Signal processing; BIDS; GUI; Known limitations (appendix, folded from `ICNNA_Limitations.doc`); Migration from legacy (`.mat` load-compat rule documented here).
5. Migration plan from `ICNNA_UserGuide.pdf` + MENA guide lineage to LaTeX.

**E.** 1–2 sessions.

**R.** Scope of LaTeX migration; only skeleton here, content migrates progressively.

### 5.2 Arc B — Data-model spine + experimental-design generalisation

Order derived from dependency graph: `signal` → `structuredData` → `dataSource` → `session` → `experiment`. `*Definition` siblings decoupled (Rule 7) except where natural pairing emerges.

#### v1.4.4 — `signal` + `structuredData` + `neuroimage` + `nirs_neuroimage`

**G.** Lowest layer of the spine; flat-architecture redesign, not port (P5).

**M.** `+icnna.data.core.signal`, `.structuredData`, `.neuroimage`, `.nirs_neuroimage`; `tests/data/core/`; `+icnna.compat/load*` adapters.

**T.**
1. Design note on `signal` / `structuredData` interface (P5).
2. Implement modern `signal`; tests.
3. Implement modern `structuredData`; tests.
4. Implement `neuroimage` and `nirs_neuroimage` as `structuredData` specialisations; tests.
5. Ship `+icnna.compat.loadSignal`, `loadStructuredData`, `loadNeuroimage`, `loadNirs_neuroimage` (N.6).
6. Deprecate `oo/@signal`, `oo/@structuredData`, `oo/@neuroimage`, `oo/@nirs_neuroimage`.

**E.** 3–4 sessions.

**R.** `structuredData` has deep semantic ties to analysis pipelines; strict P9.

#### v1.4.5 — `dataSource` + `dataSourceDefinition`; retire `@rawData_Snirf`

**G.** Mid-spine layer; resolve HC6 (D.16).

**M.** `+icnna.data.core.dataSource`, `.dataSourceDefinition`; `oo/@rawData_Snirf` retirement; `+icnna.compat/load*`.

**T.**
1. Design note on `dataSource` ↔ `dataSourceDefinition` interface (paired by necessity).
2. Implement modern `dataSource`; tests.
3. Implement modern `dataSourceDefinition`; tests.

4. D.16: migrate remaining `oo/@rawData_Snirf` callers to `+icnna.data.snirf.*`; retire the legacy folder.
5. Ship `loadDataSource`, `loadDataSourceDefinition` (N.6).
6. Deprecate `oo/@dataSource`, `oo/@dataSourceDefinition`.
7. Add `+icnna.compat.toModernDataSource` if $\beta$-bridging needed.

**E.** 2–3 sessions.

**R.** Pipeline analyses may assume old `dataSource` shape.

#### v1.4.6 — `session` + `sessionDefinition`

**G.** Session layer; holds modern `dataSource` by ID (flat).

**M.** `+icnna.data.core.session`, `.sessionDefinition`.

**T.**
1. Design note (paired).
2. Implement modern `session` with multi-dataSource support; tests.
3. Implement modern `sessionDefinition` (or defer per Rule 7).
4. Ship `loadSession`, `loadSessionDefinition` (N.6).
5. Deprecate legacy; compat adapters as needed.

**E.** 2 sessions.

**R.** Resist folding experimental-design generalisation here — that is v1.4.7.

#### v1.4.7a — Experimental-design & experiment-container design memo

Design-only release per P5. **Requires FOE input**: Cantor-era pre-collaboration notes on experimental designs.

**G.** Agree on the generalised experiment model before implementing it.

**M.** `doc/designMemos/experimentModel.tex`; `doc/UML/plantuml/experimentModel.puml`; `claude/minutes/`

**T.**
1. Read and extract FOE's pre-collaboration Cantor notes.
2. Enumerate 5–8 concrete design scenarios ICNNA currently fails cleanly (event-related, within-group allocation, multifactorial, etc.).
3. Design memo: UML of revised `experiment`/`experimentalGroup`/`experimentalUnit`/`condition`/`tim` relationships; mapping each scenario → revised model; non-goals (what stays Cantor-side).
4. Record decisions in `CHANGELOG.md` (Unreleased) and project memory.

**E.** 1–2 sessions.

**R.** Gated on FOE material; advance with placeholders if delayed.

#### v1.4.7b — `experiment` redesign implementation

**G.** Implement the v1.4.7a memo; establish performance benchmark baseline.

**M.** `+icnna.data.core.experiment`; possibly extended `experimentalGroup`/`experimentalUnit`/`condition`; `tests/performance/`.

**T.**
1. Implement modern `experiment` per memo; holds IDs, not nested objects.
2. Extend design-support classes per memo if needed.
3. Full test suite for identified design scenarios.

4. Ship `loadExperiment` (N.6).
5. Migration notes in `CHANGELOG.md` under Added/Changed.
6. Deprecate `oo/@experiment`.
7. **Performance benchmark harness baseline** (N.7): `tests/performance/` with 3–5 representative workflows on fixture data; baseline committed. See §6 for the pass-by-value clarification.

**E.** 3–4 sessions.

**R.** Largest single redesign of the plan; may split v1.4.7b-i (core `experiment`) / -ii (design-support classes).

#### v1.4.8 — Spine cleanup + UML refresh + User Guide Data-model chapter

**G.** Drain compat ledger for spine-layer; refresh UML; first User Guide content landing.

**T.**
1. Audit `+icnna.compat/`; retire adapters whose legacy class is now unreachable.
2. Caller-migration sweep: any legacy spine class with caller count $\leq\sim 3$ (D.1 threshold, tentatively 3; confirm during sweep) — migrate & delete.
3. Full `m2uml` refresh of modern spine; commit under `doc/UML/ICNNA/v1.4.8/`.
4. Update PlantUML flat-architecture diagram.
5. **User Guide chapter**: "Data model" migrated to LaTeX (first chapter land, T.1 thread).
6. Benchmark re-run against v1.4.7b baseline.
7. Release notes + CHANGELOG.

**E.** 1–2 sessions.

**R.** Low; designed as a breather release.

### 5.3 Arc C — Peripheral modernisation + behaviour

#### v1.4.9 — `rawData` base audit & close

**G.** Confirm modern `rawData` base is complete; lock interface before subtype family.

**T.** audit `+icnna.data.core.rawData`; add missing tests; interface-freeze note.

**E.** 1 session.

**R.** Low.

#### v1.5.0 — rawData subtype family

**G.** Modernise hardware-specific subtypes.

**M.** `+icnna.data.misc.rawData_ETG4000`, `_NIRScout`, `_Shimadzu`, `_UCLWireless`, `_LSL`, `_BioHarnessECG` (following `+icnna.data.misc.rawData_NIRx` convention).

**T.** one subtype per 0.5–1 session; tests with small vendor fixture files; deprecate legacy; `loadRawData_*` adapters per subtype (N.6); flag fixture provenance (T.8).

**E.** 3–4 sessions (may split).

**R.** Vendor-format drift; mitigation fixture-based regression tests.

### v1.5.1 — Spatial/ROI layer

**G.** Spatial-infrastructure classes that registration depends on.

**M.** `+icnna.data.core.channelLocationMap`, `.cluster`, `.imagePartition`, `.roi`, grids (`.logarithmicRadial` `.menaGrid`).

**T.** modernise each; invariant-based tests; deprecate legacy; `+icnna.compat/load*`; migrate modern callers.

**E.** 3 sessions.

**R.** Low current use ⇒ low-coverage tests; mitigation invariant-based tests.

### v1.5.2 — Registration I: standards mapping

**G.** Modernise registration computation layer aligned to standards (10/20, 10/10, 10/5).

**M.** `+icnna.op.registration.*`; `+icnna.data.core.referenceLocation` (if not modern).

**T.**
1. Design note on registration API surface (separation from GUI).
2. Implement standards-system mapping (10/20 family); tests with known optode layouts.
3. Modernise mesh-related helpers driven by `channelLocationMap`.
4. Distance and projection primitives.

**E.** 2–3 sessions.

**R.** Research-drift absorption (handled at v1.5.3).

### v1.5.3 — Registration II: atlas alignment + research absorption

**Requires FOE input**: research material on atlas alignment.

**G.** Absorb FOE-provided research on atlas alignment into the registration API.

**T.**
1. Read and extract the research material.
2. Design memo on atlas-alignment API (`doc/designMemos/registration.tex`).
3. Implement atlas-alignment operations; tests on synthetic + known atlases.

**E.** 2–3 sessions (may split v1.5.3a/b once material reviewed).

**R.** Scope uncertainty; memo-first controls it.

### v1.5.4 — QC behaviour (survival tier)

**G.** Give `+icnna.data.core.qc/qcCriterion` concrete behaviour for all criteria currently in legacy ICNNA.

**T.**
1. Inventory current ICNNA criteria (saturation, mirroring, channel integrity, ...).
2. Define abstract `qcCriterion.evaluate(data)` → `qcResult` contract.
3. Implement concrete criteria: `qcCriterion_Saturation`, `_Mirroring`, `_ChannelIntegrity`, ...(one per legacy check).
4. `qc` aggregation + reporting interface.
5. Tests per criterion against fixture data.

**E.** 2–3 sessions.

**R.** Survival criteria assumed stable; enhancement criteria deferred to v1.6.4.

**v1.5.5 — `analysis` + `experimentSpace` + `integrityStatus` + `ecg` + UML refresh + User Guide Analysis chapter**

**G.** Top of analysis chain; second User Guide chapter; peripheral UML refresh.

**T.**
1. Design note on `analysis`/`experimentSpace` interface (P5).
2. Modernise each class; tests.
3. Migrate callers; deprecate legacy; `+icnna.compat/load*`.
4. `m2uml` peripheral refresh.
5. **User Guide chapter**: "Analysis" migrated/updated.
6. Benchmark re-run.

**E.** 3 sessions.

**R.** Moderate; `experimentSpace` semantics may need clarification — contain via memo.

**v1.6.0 — Signal processing I: survival-tier core ops**

**G.** Essential DSP ops expected by current users; missing or scattered today. Minimise Signal Processing Toolbox dependency.

**M.** `+icnna.op.signal.*`.

**T.**
1. Bandpass / high-pass / low-pass filters (FIR + IIR) via base MATLAB `filter` where feasible; coefficient design via simple analytic formulae or offline. Flag if circumvention proves impractical.
2. Detrending (polynomial, moving average).
3. Concentration conversions audit: move existing `deltaOD2deltaConcentrations`, `intensity2deltaOD` `invertMBLL` to `+icnna.op.signal.mbll.*`.
4. DC removal, baseline correction.
5. Tests: impulse/step responses; dimensional consistency per global CLAUDE.md §2.5.

**E.** 3 sessions.

**R.** Algorithm correctness; mitigation impulse/step-response tests.

**v1.6.1 — Signal processing II: motion correction beyond TDDR**

**G.** Broader motion-correction toolkit; optode-movement is a stated concern.

**M.** `+icnna.op.signal.motion.*`.

**T.**
1. Survey: TDDR (existing), spline interpolation, wavelet-based, CBSI, hybrid. Design note.
2. Implement top 2–3 per survey recommendation.
3. Unified motion-correction interface (method selection + composition).
4. Tests with synthetic movement artefacts (ties to v1.6.3).

**E.** 2–3 sessions.

**R.** Method selection; mitigation unified interface, user picks.

### v1.6.2 — Signal processing III: physiological regression (SPA-like and beyond)

**G.** Physiological-regression toolkit beyond short-channel.

**M.** `+icnna.op.signal.physioReg.*`.

**T.**
1. Design note: short-channel regression, SPA, adaptive filtering, ICA-based approaches.
2. Implement short-channel regression (baseline).
3. Implement SPA-like regression.
4. Tests with synthetic data (ties to v1.6.3).

**E.** 2–3 sessions.

**R.** Research drift; expect refinement cycles.

### v1.6.3 — Synthetic data generation + reproducibility policy

**G.** Generate synthetic fNIRS data with known ground truth for testing + method validation.

**M.** `+icnna.op.synthesis.*`.

**T.**
1. Design note on synthetic data model: HRF shapes, noise models, physiological contamination (Mayer, respiration, cardiac), motion-artefact injection.
2. Implement HRF-based signal generator.
3. Implement noise generators (white, pink, physiological).
4. Implement motion-artefact injection to validate v1.6.1.
5. Round-trip tests through modernised pipeline.
6. **Random-seed policy**: all synthesis ops accept optional seed; default reproducible; `doc/conventions/reproducibility.tex`.

**E.** 2–3 sessions.

**R.** Research-grade scope; mitigation minimum-first, iterate.

### v1.6.4 — QC behaviour (enhancement tier) + User Guide QC chapter

**G.** Modern QC criteria beyond legacy.

**T.**
1. Implement `qcCriterion_SCI` (scalp coupling index).
2. Implement `qcCriterion_CV` (coefficient of variation).
3. Additional criteria from overprocessing / pipeline-optimisation literature.
4. Tests; `m2uml` refresh of QC surface.
5. **User Guide chapter**: "Quality control".

**E.** 2 sessions.

**R.** Enhancement-tier (P7); droppable to Cantor if under pressure.

### v1.7.0 — BIDS compatibility

**G.** BIDS-fNIRS export; round-trip is enhancement (D.14).

**M.** `+icnna.io.bids.*`.

**T.**
1. Pin to dated BIDS-fNIRS spec revision.
2. Implement BIDS export from modern `experiment/session/dataSource/rawData`.
3. Tests against BIDS-validator fixtures.
4. Enhancement: import (round-trip).

**E.** 2–3 sessions.

**R.** R.FM4 spec drift; mitigated by dated pin.

## 5.4 Arc D — Endpoint

### v1.8.0 — `oo/` removal sweep

**G.** Delete the legacy `oo/` tree; `+icnna.compat/` shrinks to only `load*` adapters + residual legitimate bridges.

**T.** final grep audit; delete unreachable `oo/@*` folders; update startup path; tests pass clean; CHANGELOG "Removed". **Load-compat adapters retained** (N.6) for user archives.

**E.** 1–2 sessions.

**R.** Surprise caller; mitigation full test-suite run pre-delete.

### v1.8.1 — Handover preparation

**G.** Draft handover artefacts per E.3 (may wait on Cantor maturity).

**T.**
1. Draft feature-parity checklist (D.5) with Cantor; may stay open.
2. Draft migration guide skeleton SNIRF/BIDS → Cantor (D.6).
3. Draft handover-moment announcement (D.9); still deferred per E.4.3 if premature.

**E.** 2 sessions.

**R.** Gated on Cantor progress; advance what is independent.

### v1.9.0 — New GUI design memo: unified application

Design-only release.

**G.** Design the unified GUI before building; subsume `guiRegistration` and other separate entry points.

**T.**
1. Design memo: single entry point; view system (experiment designer, registration, integrity/QC, analysis, visualisation); MVC vs MVP (decision + rationale); App Designer acceptable only with full code control (L2).
2. PlantUML component + sequence diagrams for main flows.
3. `doc/designMemos/guiArchitecture.tex`.

**E.** 1–2 sessions.

**R.** Architectural weight; strict review before v1.9.1.

### v1.9.1 — GUI I: shell + core editing

**G.** Unified GUI shell with editors for core entities.

**T.** main window + navigation; editors for experiment, subject, session, dataSource; tests where feasible.

**E.** 3–4 sessions.

**R.** GUI automated-testability limits; state explicitly per user instructions.

**v1.9.2 — GUI II: analysis + visualisation views**

**G.** Integrate `+icnna.plot.*`; analysis configuration + execution view; results visualisation.

**E.** 3–4 sessions.

**R.** Scope creep from visualisation wishlist; strict P9.

**v1.9.3 — GUI III: registration + QC + integrity views**

**G.** Registration as a unified-GUI view (no separate entry point); QC dashboard; integrity view.

**E.** 3 sessions.

**R.** Registration view is novel surface; validate on one real dataset.

**v2.0.0 — Release**

**G.** Endpoint per Phase 2c §v2.0.0.

**T.**
1. Delete `+icnna.compat/` (scheduled per E.1 Q3.3); `.mat load*` adapters removed alongside.
2. Final `m2uml` + PlantUML refresh of full modern surface.
3. **User Guide LaTeX polish**: all chapters current; replace `.docx` lineage.
4. README post-handover acknowledgement line (D.10) if handover has occurred; otherwise lands at whichever release coincides with handover.
5. Version tag, stable zip, release notes.

**E.** 3–4 sessions.

**R.** Final-polish drift; set hard scope.

# 6 Performance Benchmark Clarification

The pass-by-value decision (Phase 2c, L3) sets a *ceiling* on achievable performance; some copying cost is unavoidable and **acknowledged as intended**. The benchmark harness introduced at v1.4.7b detects *regressions within that ceiling*: a redesigned class or caller that inadvertently triggers additional copies, loses a cache, or degrades from $O(n \log n)$ to $O(n^2)$. Those are redesign-introduced inefficiencies orthogonal to pass-by-value itself. The flat-architecture redesign at v1.4.7b is the release most likely to introduce such regressions, which is why the baseline lands there. Benchmarks are human-read, not CI-gated; the harness is deliberately modest.

# 7 Architectural Evolution Narrative

Arc A lays infrastructure that every later arc consumes: tests, bridging, disclosure artefacts, error-ID and toolbox conventions, CI, bug-tracking discipline, and the documentation toolchain (User Guide LaTeX skeleton, PlantUML, `m2uml`; ArgoUML retired to historical archive). The three-way split v1.4.3.1/.2/.3 reflects the weight of Arc A without violating P3.

Arc B carries the modernisation's structural thesis. The spine becomes flat and ID-referenced (P5), but v1.4.7 is explicitly a *generalisation*, not a translation: `experiment` must express event-related designs, within-group allocation, and multifactorial designs without "abusing" `session`. The memo-first split (v1.4.7a design, v1.4.7b implementation) separates the research-like design decision from its implementation. The performance benchmark harness lands at v1.4.7b, where regressions are most likely.

Arc C is where ICNNA picks up *behaviour* it never had or needs to regain. Registration takes two releases and expects research drift (FOE material). QC gains concrete criteria with a clean tier split (survival at v1.5.4, enhancement at v1.6.4). Signal processing is a three-release arc (v1.6.0–v1.6.2) covering core ops, motion correction, and physiological regression. Synthetic data (v1.6.3) doubles as validation infrastructure for v1.6.1 and v1.6.2. BIDS at v1.7.0 completes the standards posture begun with SNIRF.

Arc D is deletion + delivery. `oo/` disappears (v1.8.0) while `.mat` load-compat adapters are retained for user archives. Handover artefacts land where feasible (v1.8.1). The unified GUI is designed before built (v1.9.0 memo, v1.9.1–v1.9.3 implementation), subsuming the registration-tool entry point. v2.0.0 ships `+icnna`-only, with a completed LaTeX User Guide replacing the `.docx` lineage.

### 7.1 Research integration points

- **v1.4.7a** (experimental designs): consumes FOE's Cantor-era design notes.
- **v1.5.3** (atlas alignment): consumes FOE-provided research material.
- **v1.6.1–v1.6.3** (motion, physiological regression, synthetic data): informed by overprocessing / pipeline-optimisation theses; may trigger new research.
- **v1.6.4** (enhancement QC): SCI, CV, and other methods from the thesis literature.

### 7.2 New research likely triggered

- Atlas-alignment method choice if existing research is incomplete.
- Synthetic-data ground-truth model — may warrant a publishable specification.
- Experimental-model generalisation — may itself be publishable as a methodology note.

## 8 Risk and Dependency Summary

| Risk | Versions | Mitigation |
|---|---|---|
| R.FM1 stuck-in-the-middle | Arc B, esp. v1.4.7 | P3; v1.4.8 catch-up; v1.4.7 memo-first split |
| R.HC2 dual-arch friction | Arc A–C | `+icnna.compat`; drains progressively; final drain v2.0.0 |
| R.HC5 research interrupts | All | Rule 2 mixed granularity; memo-first on research-heavy releases |
| R.F1 lag ICNNA/Cantor | Arc C–D | Claude standing responsibility to flag |
| R.F2 single-developer bus factor | All | Sequenced: front-load ICNNA, then Cantor primary |
| R.FM4 BIDS drift | v1.7.0 | Pin to dated spec; tests as floor |
| D.7 IP ledger blocked | v1.4.3.1 | Placeholder, advance, revisit |
| Research drift registration | v1.5.2, v1.5.3 | Memo-first; splittable |
| Research drift signal proc | v1.6.1–v1.6.3 | Synthetic data (v1.6.3) provides validation floor |

| Risk | Versions | Mitigation |
|---|---|---|
| GUI design weight | v1.9.0 | Design-only release before implementation |
| User `.mat` archive breakage | v1.8.0 | Load-compat adapters retained past deletion until v2.0.0 |
| Performance regression | v1.4.7b onwards | Benchmark harness at v1.4.7b; re-run each Arc-B release |
| User Guide scope | Cross-cutting | Chapter-by-chapter landings; no big-bang |

### 8.1 Critical path

v1.4.2 → v1.4.3.* → Arc B spine → v1.4.8 cleanup → v1.8.0 `oo/` removal → v2.0.0

Arc C releases can reorder under research interrupt without violating the critical path. Arc D handover artefacts (v1.8.1) gate on Cantor progress for D.5/D.6 but not for the critical path to v2.0.0.

### 8.2 Session budget summary

| Arc | Sub-releases | Sessions (low) | Sessions (high) |
|---|---|---|---|
| A | 4 | 6 | 8 |
| B | 7 | 12 | 17 |
| C | 13 | 27 | 39 |
| D | 7 | 15 | 25 |
| **Total** | **31**[1] | **60** | **89** |

## 9 Decisions Recorded

| Area | Decision |
|---|---|
| Sub-release count | 28 at roadmap level; up to 31 if v1.4.7b and v1.5.3 splits fire. |
| Session budget | ∼60–89 sessions total over ∼2–3 years at research pace. |
| Arc structure | A Readiness / B Spine+Experimental-design / C Peripheral+Behaviour / D Endpoint. |
| v1.4.3 split | Three-way into v1.4.3.1 Disclosure / v1.4.3.2 Conventions+tooling / v1.4.3.3 Documentation scaffolding. |
| Cross-cutting threads | Eight (User Guide, UML, compat drain, `.mat` load-compat, performance benchmarks, examples cookbook, bug tracking, fixture provenance). |
| Experimental-design generalisation | Survival-tier; memo-first at v1.4.7a; implementation v1.4.7b. Requires FOE notes. |
| Registration as first-class | Two-release arc v1.5.2 (standards) + v1.5.3 (atlas + research). Requires FOE research material. |
| QC tiering | Survival tier v1.5.4 (current ICNNA criteria); enhancement tier v1.6.4 (SCI, CV, etc.). |

[1] Counting v1.4.3.1/.2/.3 as three sub-releases and allowing potential v1.4.7b-i/-ii and v1.5.3a/b splits. Headline count is 28 at the roadmap level.

| Area | Decision |
|---|---|
| Signal processing | Three-release arc v1.6.0 survival / v1.6.1 motion / v1.6.2 physiological; synthetic data v1.6.3. |
| Signal Processing Toolbox | Avoid where feasible; Statistics Toolbox accepted. |
| UML toolchain | Keep `m2uml`; adopt PlantUML; retire ArgoUML to `doc/UML/historical/`. |
| User Guide | LaTeX migration from `.docx` lineage; skeleton v1.4.3.3; chapter landings v1.4.8/v1.5.5/v1.6.4; polish v2.0.0. Fold `ICNNA_Limitations.doc` as appendix. |
| `.mat` load-compat | Per-class adapter in `+icnna.compat.load<ClassName>.m`; retained past v1.8.0; removed v2.0.0. |
| Performance benchmarks | Baseline v1.4.7b; re-run each Arc-B release; human-read; pass-by-value ceiling acknowledged as intended. |
| Error-ID convention | Codify current `icnna:<namespace>:<reason>` style at v1.4.3.2; no behavioural change. |
| MATLAB toolbox deps | Statistics: accepted. Signal Processing: avoid where feasible; declare at v1.4.3.2. |
| Citation | `CITATION.cff` at v1.4.3.1 using DOI 10.1117/1.NPh.5.1.011011. |
| CI | Hybrid: GitHub Actions for `matlab.unittest` + lint on PR/-push; coverage local; landed at v1.4.3.2. |
| Bug tracking | GitHub Issues with labels; `CONTRIBUTING.md`; User Guide "Known limitations" appendix. |
| GUI at v2.0.0 | Unified application subsuming `guiRegistration` and other separate entry points. Design memo v1.9.0; implementation v1.9.1–.3. |
| D.16 `@rawData_Snirf` | Retired at v1.4.5 with `dataSource` modernisation. |
| D.1 cleanup threshold | Tentatively $\leq 3$ callers; confirm at v1.4.8 sweep. |
| D.14 BIDS scope | Export survival-tier; round-trip enhancement. |

# 10 Slotting of Phase 2d Deferred Decisions

# 11 Hand-off to Phase 4

**Phase 4 objective:** execution planning for v1.4.2 — session-level breakdown of the six v1.4.2 tasks into a concrete one-to-two-session plan; acceptance criteria per task; test-harness structure decision (`matlab.unittest` suite layout, fixture directory layout); GUI deprecation-banner text; `+icnna.compat.toModernTimeline` implementation skeleton.

**Phase 4 inputs.**

- This minute, especially §v1.4.2.
- Phase 2d minute §E.1 Q4 (v1.4.2 pre-packed content).
- Phase 2c minute §P1–P9 (design principles).
- The `ROADMAP.md` at repo root.

**Phase 4 starting action.** FOE will provide a Phase 4 context prompt at session start. A draft prompt will be produced at `claude/prompts/Phase4_context_prompt.txt` as fallback.

| ID | Version | Note |
|---|---|---|
| D.1 | v1.4.8 | Cleanup-trigger threshold confirmed at sweep |
| D.2 | v1.4.3.2 | Deprecation-warning mechanism convention |
| D.3 | v1.4.3.2 | Doxygen header template for `+icnna.compat` |
| D.4 | v1.4.8 (tentative) | ICNNA→Cantor priority-shift milestone; revisited per Claude lag-flag |
| D.5 | v1.8.1 | Feature-parity checklist; may stay open |
| D.6 | v1.8.1 | Migration guide SNIRF/BIDS → Cantor |
| D.7 | v1.4.3.1 | IP ledger collaborative pass |
| D.8 | v1.4.3.2 | MATLAB-version support statement phrasing |
| D.9 | v1.8.1 (draft) / deferred | Handover announcement draft; form/channels/timing still deferred |
| D.10 | v2.0.0 (or handover release) | Post-handover README acknowledgement wording |
| D.11 | v1.4.3.1 | CHANGELOG seeding (full migration, §9) |
| D.12 | v1.4.2 (seed) / cross-cutting cookbook | Worked-example content; gallery grows organically |
| D.13 | v1.9.0 | GUI v2.0.0 design memo |
| D.14 | v1.7.0 | BIDS compatibility |
| D.15 | v1.4.2 | Test infrastructure (`matlab.unittest`) |
| D.16 | v1.4.5 | `@rawData_Snirf` retirement with `dataSource` |

# 12 Phase 3 Closing Statement

## 12.1 What Phase 3 delivered

1. A concrete versioned roadmap from v1.4.1-beta to v2.0.0 comprising 28 sub-releases across four arcs.
2. All 16 Phase 2d deferred decisions slotted.
3. Seven additional Phase 3 items (N.1–N.7) folded into the roadmap.
4. Eight cross-cutting threads defined.
5. A session budget estimate (∼60–89 sessions) calibrating expectations over the ∼2–3-year horizon.
6. A `ROADMAP.md` at the repository root (committed with this minute) providing the public-facing version of the plan.
7. Handoff to Phase 4 (execution planning for v1.4.2).

## 12.2 What Phase 3 did not decide

- Session-level task breakdown within any sub-release — that is Phase 4 for v1.4.2 and per-version execution-planning sessions thereafter.
- Specific test framework specifics beyond `matlab.unittest` — Phase 4.
- Implementation detail of any modernised class — per-release design-memo or implementation session.
- Exact cleanup-threshold numeric for D.1 — confirmed at v1.4.8 when sweep runs.
- Signal-processing method shortlists (v1.6.1, v1.6.2) — per-release survey notes.

## 12.3 Sufficiency for a cold start

Phase 3 closes with the explicit intent that Phase 4 can begin from a cold context on any device, reading only the canonical artefacts under `claude/` plus `ROADMAP.md`. The portability validation established at Phase 3 start (FOE's prompt + this minute's slotting) reconfirms the canonical-context design works.

## 13 Documents and Artefacts Referenced

1. **Phase predecessors.** `claude/minutes/Phase{1,2a,2b,2c,2d}_minute.{tex,pdf}`.
2. **Global and project rules.** Global `CLAUDE.md` (§§8–9 in particular); project `CLAUDE.md`.
3. **Canonical memory.** `claude/memory/MEMORY.md` and referenced entries; new entry `claude/memory/proj` added.
4. **This roadmap's public artefacts.** `ROADMAP.md` (repo root); `CHANGELOG.md` (v1.4.3.1 deliverable, not yet present); `CITATION.cff` (v1.4.3.1 deliverable).
5. **Phase 4 handoff.** `claude/prompts/Phase4_context_prompt.txt` (draft).
6. **Cantor project location.** `C:\Users\orihuelf-admin\OneDrive\Git\Cantor`.